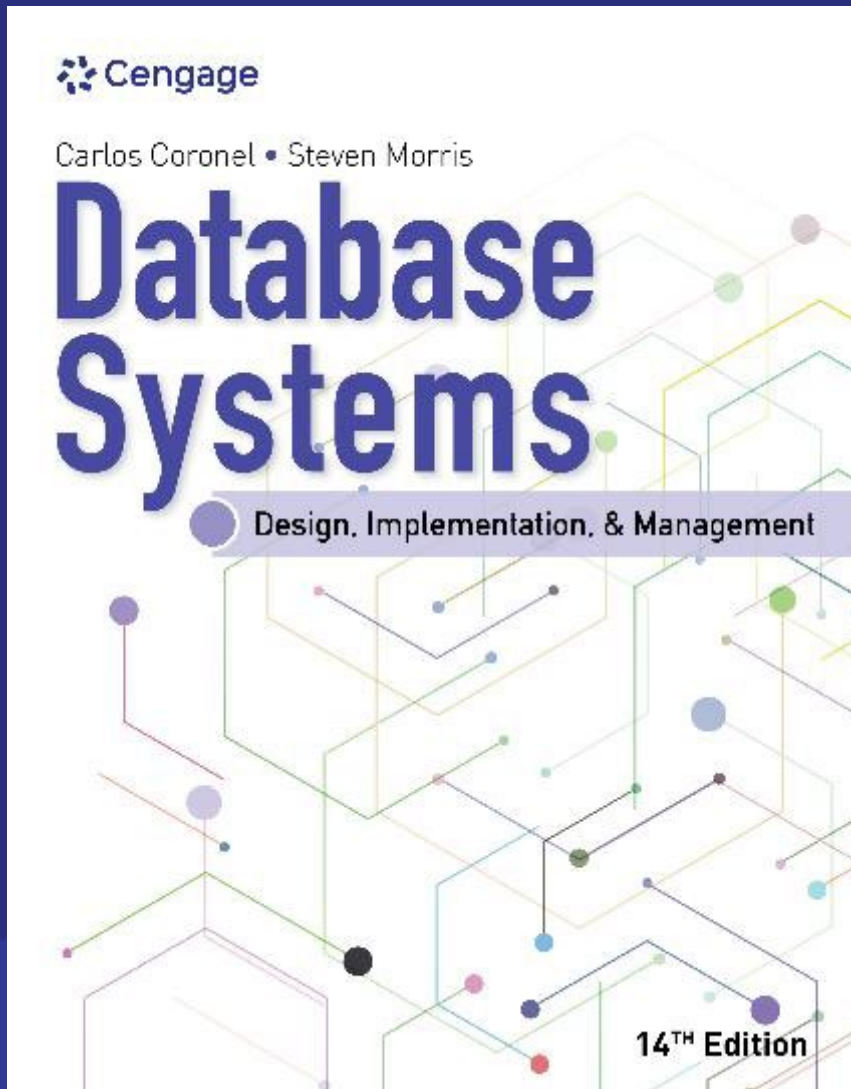




Database Systems: Design, Implementation, and Management, 14e

Module 3: The Relational Database Model

Today's Topics

- Logical structure of relational data
- Tables, attributes, domains, and rows
- Keys and functional dependencies
- Entity and referential integrity
- Relationships and the ER model
- Data redundancy and foreign keys

A Logical View of Data

- The relational model represents data logically using tables and relationships.
- Designers can focus on what the data means and how it is related, rather than how it is physically stored.
- This logical simplicity makes relational databases easier to design, understand, and query.

Tables and Their Characteristics

- A table (relation) is a two-dimensional structure made of rows and columns.
 - A row (tuple) represents one occurrence of an entity.
 - A column (attribute) represents a characteristic of that entity.
 - A domain defines the allowable values for an attribute.
 - A primary key (PK) uniquely identifies each row in a table.

Tables and Their Characteristics (2 of 2)

Table 3.1	Characteristics of a Relational Table
1	A table is perceived as a two-dimensional structure composed of rows and columns
2	Each table row (tuple) represents a single entity occurrence within the entity set
3	Each table column represents an attribute, and each column has a distinct name
4	Each intersection of a row and column represents a single data value
5	All values in a column must conform to the same data format
6	Each column has a specific range of values known as the attribute domain
7	The order of the rows and columns is immaterial to the DBMS
8	Each table must have an attribute or combination of attributes that uniquely identifies each row

Why Keys Matter

- A key consists of one or more attributes used to identify rows or connect tables.
 - Keys help ensure that each row can be uniquely identified.
 - Keys establish relationships between tables.
 - Keys support data integrity and help prevent inconsistent data.

Functional Dependencies

- A functional dependency exists when the value of one attribute (or set of attributes) determines another attribute.
 - The attribute that determines another value is called the determinant.
 - The attribute whose value is determined is called the dependent.
 - Example: $\text{StudentID} \rightarrow \text{StudentName}$, meaning each StudentID identifies one StudentName.

Composite Keys and NULL Values

- A composite key is a key made of two or more attributes.
 - An attribute that is part of a key is called a key attribute.
 - NULL represents the absence of a data value.
 - No part of a primary key may be NULL.
 - A NULL may represent an unknown, missing, or not-applicable value.

Foreign Keys and Referential Integrity

- A foreign key (FK) is an attribute in one table that refers to the primary key of another table.
- Referential integrity requires each non-NULL foreign key value to match an existing primary key value in the related table.
 - This prevents invalid references between related tables.

Types of Keys (3 of 3)

Table name: **PRODUCT**
Primary key: **PROD_CODE**
Foreign key: **VEND_CODE**

Database name: Ch03_SaleCo

PROD_CODE	PROD_DESCRIPTOR	PROD_PRICE	PROD_ON_HAND	VEND_CODE
001278-AB	Claw hammer	12.95	23	232
123-21UUY	Houselite chain saw, 16-in. bar	189.99	4	235
QER-34256	Sledge hammer, 16-lb. head	18.63	6	231
SRE-657UG	Rat-tail file	2.99	15	232
ZZX/3245Q	Steel tape, 12-ft. length	6.79	8	235

link

Table name: **VENDOR**
Primary key: **VEND_CODE**
Foreign key: none

VEND_CODE	VEND_CONTACT	VEND_AREACODE	VEND_PHONE
230	Shelly K. Smithson	608	555-1234
231	James Johnson	615	123-4536
232	Annelise Crystall	608	224-2134
233	Candice Wallace	904	342-6567
234	Arthur Jones	615	123-3324
235	Henry Ortozo	615	899-3425

Figure 3.2 An Example of a Simple Relational Database

Knowledge Check

- **What is the difference between a database and a table?**

Knowledge Check: Answer

- A table is one component of a database. It stores data about a specific entity or subject.
- A database contains one or more related tables plus metadata that describes the data and relationships.

Integrity Rules (1 of 2)

Table 3.4	Integrity Rules
Entity Integrity	Description
Requirement	All primary key entries are unique, and no part of a primary key may be null.
Purpose	Each row will have a known, unique identity, and foreign key values can properly reference primary key values.
Example	No invoice can have a duplicate number, nor can it be null; in short, all invoices are uniquely identified by their invoice number.
Referential Integrity	Description
Requirement	A foreign key may have either a null entry, as long as it's not part of its table's primary key, or an entry that matches the primary key value in a table to which it is related (every non-null foreign key value must reference an existing primary key value).
Purpose	The purpose is to ensure that every reference by a foreign key is a valid reference to the related primary key. It is possible for an attribute not to have a corresponding value, but it will be impossible to have an invalid entry; the enforcement of the referential integrity rule makes it impossible to delete a row in one table whose primary key has mandatory matching foreign key values in another table.
Example	A customer might not yet have an assigned sales representative (number), but it will be impossible to have an invalid sales representative (number).

Integrity Rules (2 of 2)

Table name: CUSTOMER

Database name: Ch03_InsureCo

Primary key: CUS_CODE

Foreign key: AGENT_CODE

CUS_CODE	CUS_LNAME	CUS_FNAME	CUS_INITIAL	CUS_RENEW_DATE	AGENT_CODE
10010	Ramas	Alfred	A	05-Apr-2024	502
10011	Dunne	Leona	K	16-Jun-2024	501
10012	Smith	Kathy	W	29-Jan-2025	502
10013	Olowski	Paul	F	14-Oct-2024	
10014	Orlando	Myron		28-Dec-2024	501
10015	O'Brian	Amy	B	22-Sep-2024	503
10016	Brown	James	G	25-Mar-2025	502
10017	Williams	George		17-Jul-2024	503
10018	Farriss	Anne	G	03-Dec-2024	501
10019	Smith	Olette	K	14-Mar-2025	503

Table name: AGENT (only five selected fields are shown)

Primary key: AGENT_CODE

Foreign key: none

AGENT_CODE	AGENT_AREACODE	AGENT_PHONE	AGENT_LNAME	AGENT_YTD_SLS
501	713	228-1249	Alby	132735.75
502	615	882-1244	Hahn	138967.35
503	615	123-5589	Okon	127093.45

Figure 3.3 An Illustration of Integrity Rules

Relationships in a Relational Database

- One-to-many (1:M): one row in one table can relate to many rows in another table. This is the most common relationship.
- One-to-one (1:1): one row relates to at most one row in another table, and vice versa.
- Many-to-many (M:N): implemented using an associative (bridge/linking) table.
 - The associative table typically contains foreign keys referencing the primary keys of the related tables.

Breaking a Many-to-Many (M:N) Relationship

- Example: STUDENT and COURSE
 - One student can take many courses, and one course can have many students.
- Create a linking (associative) table called ENROLLMENT.
 - ENROLLMENT stores StudentID and CourseID as foreign keys.
- This changes one M:N relationship into two 1:M relationships:
 - STUDENT 1:M ENROLLMENT and COURSE 1:M ENROLLMENT
- Example rows: (101, DB101), (101, PR102), (102, DB101)

The Entity Relationship Model

- The Entity Relationship (ER) model provides a graphical way to represent data requirements.
 - An Entity Relationship Diagram (ERD) shows entities, attributes, and relationships.
 - The ER model and relational model work together as the foundation of database design.

Core ER Model Components

- **Entity:** a person, place, object, event, or concept about which data is stored.
- **Attribute:** a characteristic that describes an entity.
- **Relationship:** an association between entities.
- Common ERD notations include Chen, Crow's Foot, and UML class diagrams.

The Entity Relationship Model

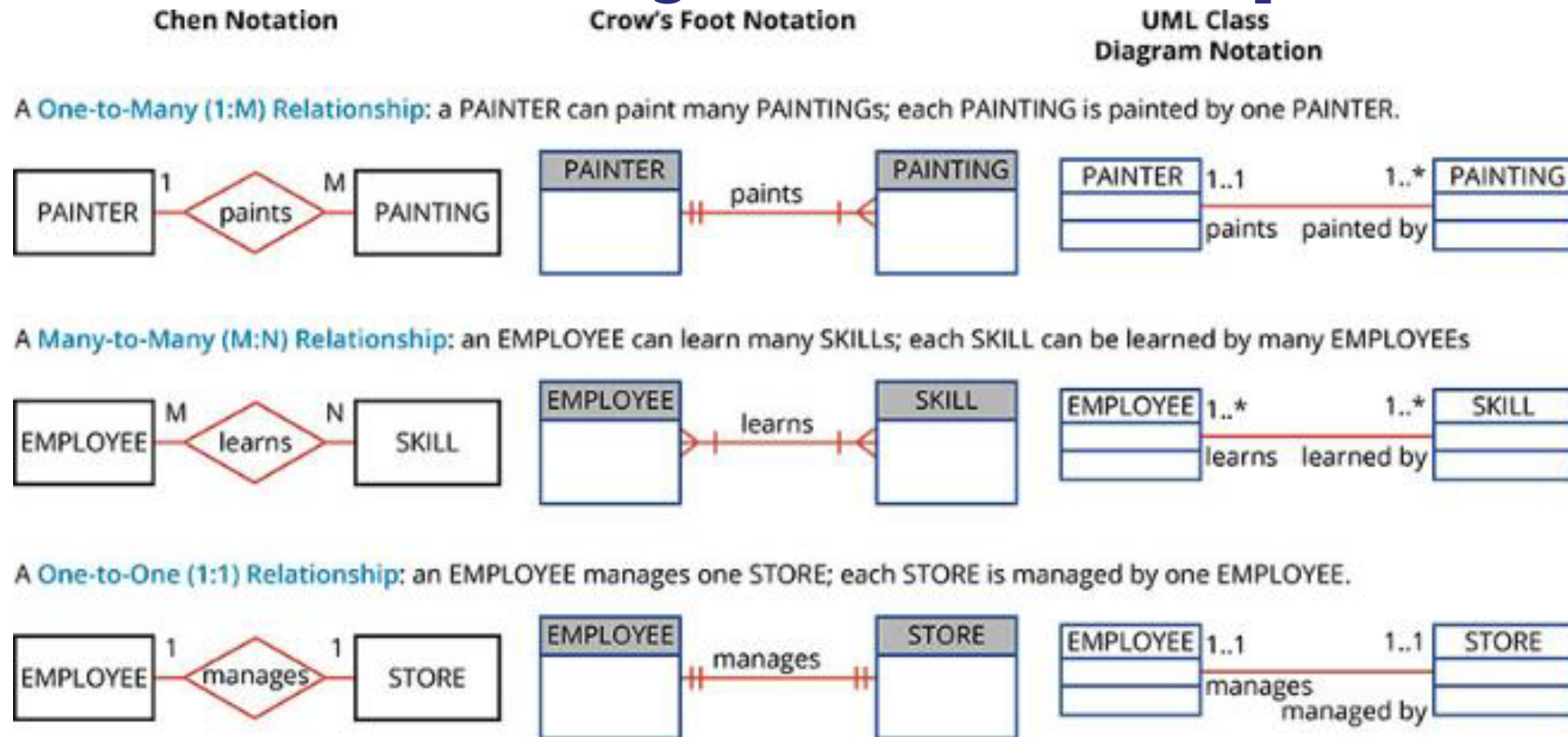
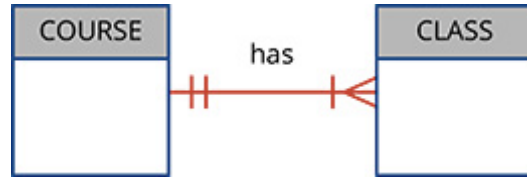


Figure 2.3 The ER Model Notations

Relationships within the Relational Database



- The 1:M Relationship between COURSE and CLASS

Table name: COURSE
Primary key: CRS_CODE
Foreign key: none

Database name: Ch03_TinyCollege

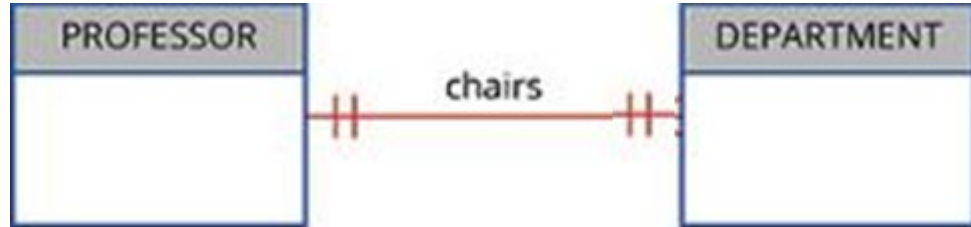
CRS_CODE	DEPT_CODE	CRS_DESCRIPTION	CRS_CREDIT
ACCT-211	ACCT	Accounting I	3
ACCT-212	ACCT	Accounting II	3
CIS-220	CIS	Intro. to Microcomputing	3
CIS-420	CIS	Database Design and Implementation	4
GM-261	CIS	Intro. to Statistics	3
GM-362	CIS	Statistical Applications	4

Table name: CLASS
Primary key: CLASS_CODE
Foreign key: CRS_CODE

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	CLASS_ROOM	PROF_NUM
10012	ACCT-211	1	MWF 8:00-8:50 a.m.	BUS311	105
10013	ACCT-211	2	MWF 9:00-9:50 a.m.	BUS300	105
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS262	342
10015	ACCT-212	1	MWF 10:00-10:50 a.m.	BUS311	301
10016	ACCT-212	2	Th 6:00-8:40 p.m.	BUS262	301
10017	CIS-220	1	MWF 9:00-9:50 a.m.	HLR209	228
10018	CIS-220	2	MWF 9:00-9:50 a.m.	HLR211	114
10019	CIS-220	3	MWF 10:00-10:50 a.m.	HLR209	228
10020	CIS-420	1	W 6:00-8:40 p.m.	HLR208	162
10021	GM-261	1	MWF 8:00-8:50 a.m.	HLR200	114
10022	GM-261	2	TTh 1:20-2:15 p.m.	HLR200	114
10023	GM-362	1	MWF 11:00-11:50 a.m.	HLR200	162
10024	GM-362	2	TTh 2:30-3:45 p.m.	HLR200	162

- The Implemented 1:M Relationship between
COURSE and CLASS

Relationships within the Relational Database



➤ The 1:1 Relationship between PROFESSOR and DEPARTMENT

Table name: PROFESSOR
Primary key: EMP_NUM
Foreign key: DEPT_CODE

Database name: Ch03_TinyCollege

EMP_NUM	DEPT_CODE	PROF_OFFICE	PROF_EXTENSION	PROF_HOURLY_RATE
101	ART	009 105	8220	Ph.D.
104	ENG	005 102	2001	MA
105	JOINT	004 2000	3085	Ph.D.
106	MATH	007 100	2000	Ph.D.
110	ENG	004 100	3012	Ph.D.
114	JOINT	007 201	4455	Ph.D.
120	MATH	004 201	1801	Ph.D.
140	ENG	005 102	2045	Ph.D.
142	ENG	005 100	2204	Ph.D.
101	MATH	004 100	2315	Ph.D.
150	JOINT	004 200	2000	Ph.D.
200	ENG	004 100	3401	Ph.D.
220	ENG	007 100	2000	Ph.D.
280	MATH	004 100	1755	Ph.D.
290	JOINT	007 201	2001	Ph.D.
301	JOINT	004 201	1801	Ph.D.
310	ENG	005 100	2000	Ph.D.
342	JOINT	004 200	3011	Ph.D.
350	ENG	004 100	3005	Ph.D.
401	ENG	005 100	2000	MA
405	JOINT	004 201	2001	MA
420	ART	009 100	2270	Ph.D.

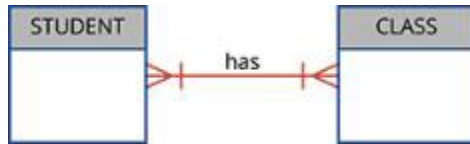
↑ The 1:M DEPARTMENT employs PROFESSOR relationship is implemented through the placement of the DEPT_CODE foreign key in the PROFESSOR table.

Table name: DEPARTMENT
Primary key: DEPT_CODE
Foreign key: EMP_NUM

↓ The 1:1 PROFESSOR chairs DEPARTMENT relationship is implemented through the placement of the EMP_NUM foreign key in the DEPARTMENT table.

DEPT_CODE	DEPT_NAME	DEPT_CODE	EMP_NUM	DEPT_EXTENSION	DEPT_HOURLY_RATE
001	Accounting	001	111	001 211	2119
002	Art	002	104	002 104	2000
003	Biological	003	201	003 201	4117
004	Chemical	004	200	004 200	2000
005	Computer Science	005	200	005 200	2000
006	Economics	006	200	006 200	2000
007	Engineering	007	100	007 100	2000
008	Health	008	100	008 100	2000
009	History	009	100	009 100	2000
010	Mathematics	010	100	010 100	2000
011	Physics	011	100	011 100	2000
012	Psychology	012	100	012 100	2000
013	Sociology	013	100	013 100	2000

Relationships within the Relational Database



➤ The ERM's M:N Relationship between STUDENT and CLASS

Table name: STUDENT
Primary key: STU_NUM
Foreign key: none

Database name: Ch03_CollegeTry

STU_NUM	STU_NAME	CLASS_CODE
321452	Bowser	10014
321452	Bowser	10018
321452	Bowser	10021
324257	Sinelson	10014
324257	Sinelson	10018
324257	Sinelson	10021

Table name: CLASS
Primary key: CLASS_CODE
Foreign key: STU_NUM

CLASS_CODE	STU_NUM	CRS_CODE	CLASS_SECTION	CLASS TIME	CLASS_ROOM	PROF_NUM
10014	321452	ACCT-211	3	TRN 2:30-3:45 pm	BUS252	342
10014	324257	ACCT-211	3	TRN 2:30-3:45 pm	BUS252	342
10018	321452	CIS-220	2	MAF 8:00-8:50 am	PLR211	114
10018	324257	CIS-220	2	MAF 8:00-8:50 am	PLR211	114
10021	321452	GM-251	1	MAF 8:00-8:50 am	PLR200	114
10021	324257	GM-251	1	MAF 8:00-8:50 am	PLR200	114

➤ The Wrong Implementation of the M:N Relationship between STUDENT and CLASS

Relationships within the Relational Database

Table name: STUDENT
Primary key: STU_NUM
Foreign key: none

STU_NUM	STU_LNAME
321452	Bowser
324257	Smithson

Database name: Ch03_CollegeTry2

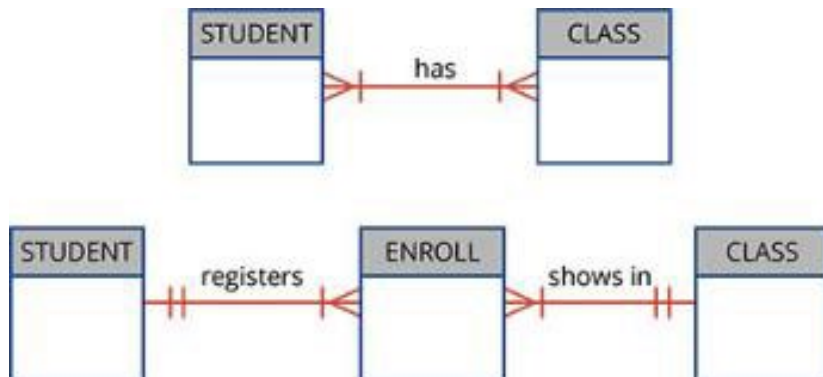
Table name: ENROLL
Primary key: CLASS_CODE + STU_NUM
Foreign key: CLASS_CODE, STU_NUM

CLASS_CODE	STU_NUM	ENROLL_GRADE
10014	321452	C
10014	324257	B
10018	321452	A
10018	324257	B
10021	321452	C
10021	324257	C

Table name: CLASS
Primary key: CLASS_CODE
Foreign key: CRS_CODE

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	CLASS_ROOM	PROF_NUM
10014	ACCT-211	3	TTh 2:30-3:45 a.m.	BLG202	342
10018	CIS-220	2	WWF 9:00-9:50 a.m.	HLR211	114
10021	GN-281	1	WWF 8:00-8:50 a.m.	HLR200	114

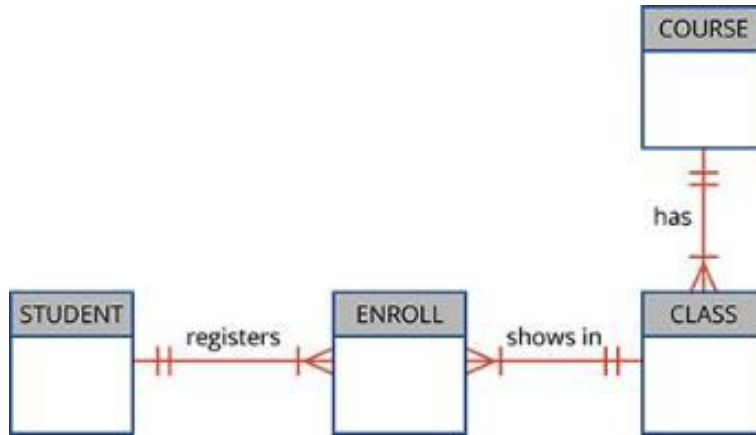
- Converting the M:N Relationship into Two 1:M Relationships
- Because the ENROLL table links two tables, it is also called a **linking table**



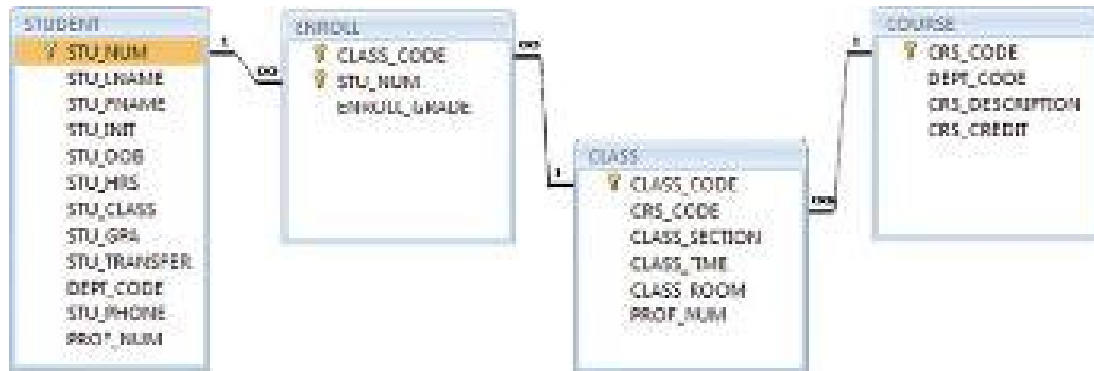
- Changing the M:N Relationships to Two 1:M Relationships

Relationships within the Relational Database

➤ The Expanded ER Model



➤ The Relational Diagram for the TinyCollege Database



Reducing Data Redundancy

- Relational databases reduce unnecessary duplication by separating data into related tables.
- Foreign keys connect related records without repeatedly storing the same descriptive data.
- Reducing redundancy improves consistency and makes updates easier to manage.

Data Redundancy Revisited (2 of 2)

Table name: CUSTOMER

Primary key: CUS_CODE

Foreign key: none

CUS_CODE	CUS_UNAME	CUS_FNAME	CUS_INITIAL	CUS_AREACODE	CUS_PHONE
10010	Samuel	Robert	A	210	904-2573
10011	Julianne	Leanne	K	210	904-1228
10012	Smith	Robert	W	210	904-2205
10013	Olson	Paul	F	210	904-2108
10014	Olson	Myron		210	202-5872
10015	Olson	Alan	B	210	904-3381
10016	Brinson	James	G	515	207-5328
10017	Albright	Conrad		515	206-2958
10018	Farrell	James	G	215	902-7155
10019	Smith	Clara	K	515	207-3609

Table name: INVOICE

Primary key: INV_NUMBER

Foreign key: CUS_CODE

INV_NUMBER	CUS_CODE	INV_DATE
1001	10014	08-Mar-22
1002	10011	08-Mar-22
1003	10012	08-Mar-22
1004	10011	08-Mar-22

Table name: PRODUCT

Primary key: PROD_CODE

Foreign key: none

PROD_CODE	PROD_DESCRIPTION	PROD_PRICE	PROD_ON_HAND	VEND_CODE
001015LAB	Clear bottles	12.00	20	255
121-21181Y	Household cleaner spray, 16.4 fl oz	8.00	4	255
001014Y01	Single bottles, 16.4 fl oz	18.00	8	254
001015Y01	Refilled kit	2.00	15	255
121-00450	Shampoo, 16.4 fl oz	8.75	8	255

Database name: Ch03_SaleCo

Table name: LINE

Primary key: INV_NUMBER + LINE_NUMBER

Foreign key: INV_NUMBER, PROD_CODE

INV_NUMBER	LINE_NUMBER	PROD_CODE	LINE_UNITS	LINE_PRICE
1001	1	121-01450	1	10.00
1001	2	001-01100	3	2.00
1002	1	001-01100	2	10.00
1002	1	121-01450	1	8.75
1002	2	001-01100	1	2.00
1002	3	001-01100	1	10.00
1003	1	001-01100	1	10.00
1004	2	001-01100	2	2.00

➤ A Small Invoicing System

➤ The Relational Diagram for the Invoicing System

